Last name _____

First name _____

LARSON—MATH 310–CLASSROOM WORKSHEET 20
Matrix Inverses.

Review

1. What is a left-inverse of a matrix A?
2. If A has a left-inverse, why do the columns of A have to be linearly independent?
3. What is a right-inverse of a matrix A?
4. If A has a right-inverse, why do the rows of A have to be linearly independent?

Matrix Inverses!

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$B = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

1. Check that B is a left-inverse of A (so $BA = I$).

2. Check that B is a right-inverse of A (so $AB = I$).
   So B is the *inverse* of A. We say A is *invertible* or *non-singular*.

3. Check that the columns of A are linearly independent.

4. Check that the rows of A are linearly independent.

5. Find A^{-1}.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

6. Let $\vec{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$. Use A^{-1} to solve $A\vec{x} = \vec{b}$. Write the equations and check your answer.

7. Now use the 2×2 inverse formula to find A^{-1}.

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

8. Check that Q is *orthogonal*. (It is enough to show that Q is square and $Q^T Q = I$).

9. Does $(A^{-1})^T = (A^T)^{-1}$? That's equivalent to $B^T = (A^T)^{-1}$. So check that $B^T A^T = I$.